

GSVA for mutil Group

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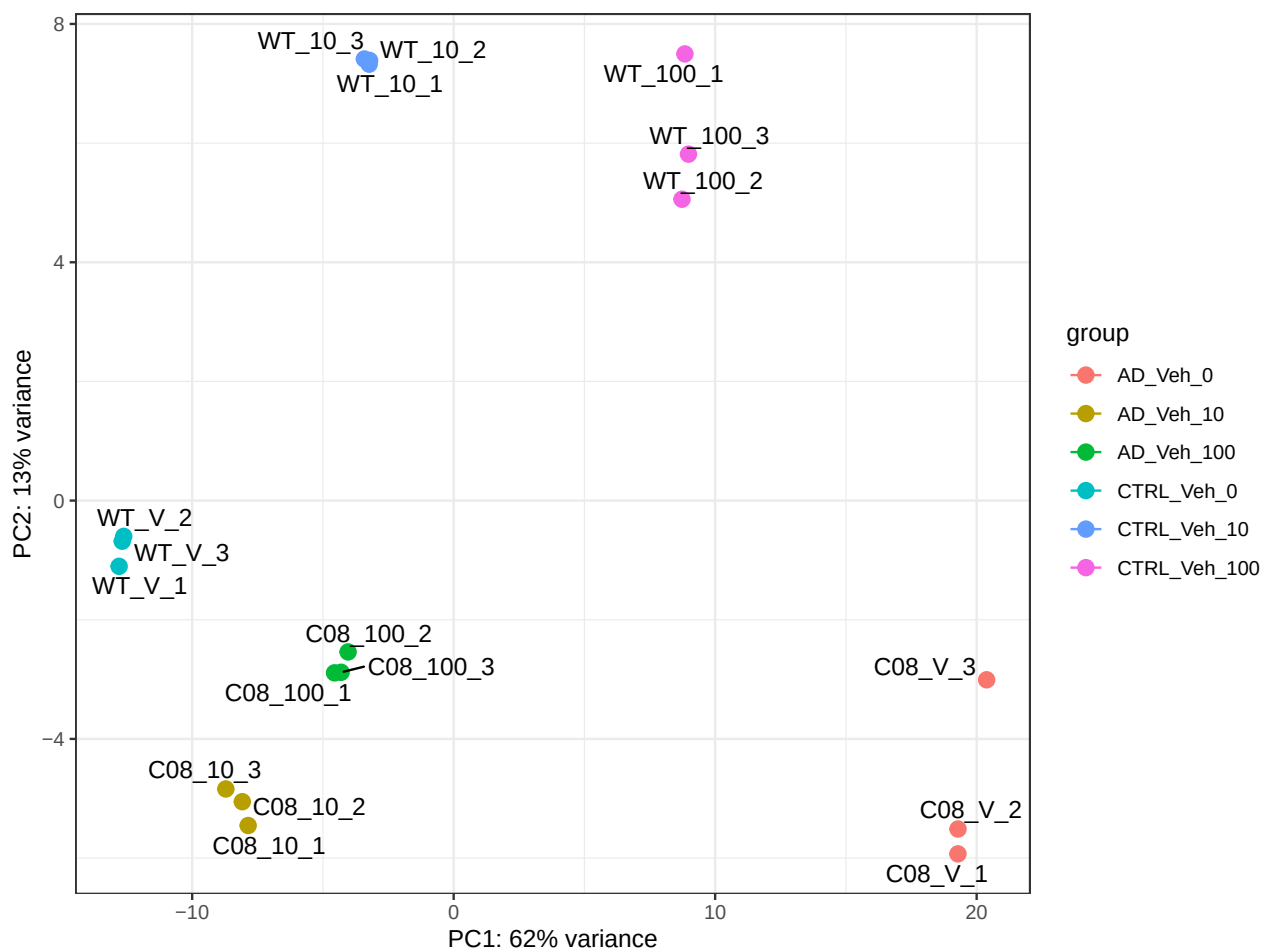
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1. Read the count data

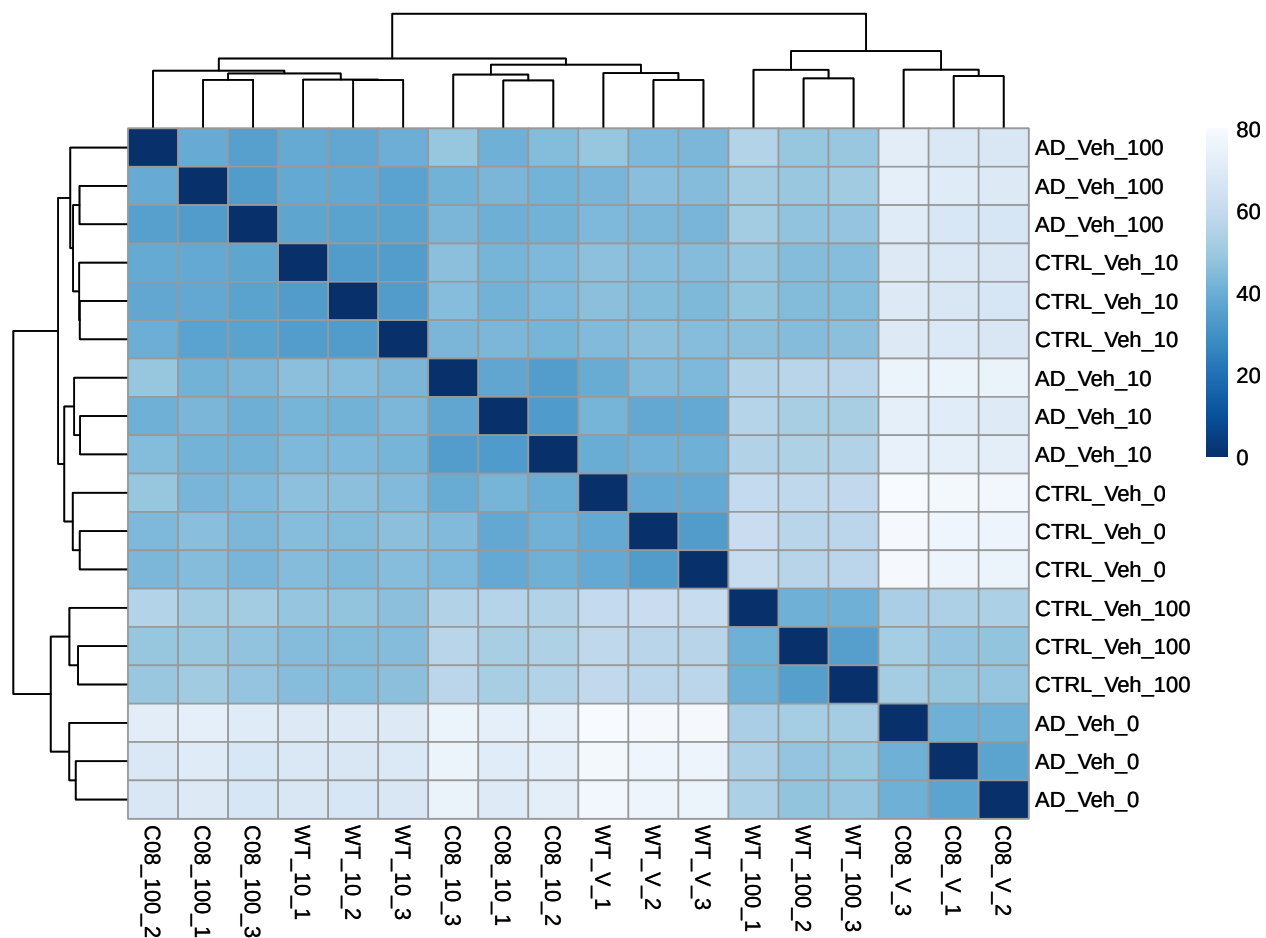
In this section, we will read the clean count data from the `synaptosomes_bulkRNA` folder. We will read the data and merge them into a single table.

2. Visualization for reuslt

(1) Sample information - PCA plot



(2) Sample information - Distance heatmap



3. GSEA analysis and BioDomain analysis

- (1) GSEA analysis for AD vs CTRL
- (2) GSEA analysis for AD_Drug vs CTRL
- (3) GSEA analysis for CTRL_Drug vs CTRL

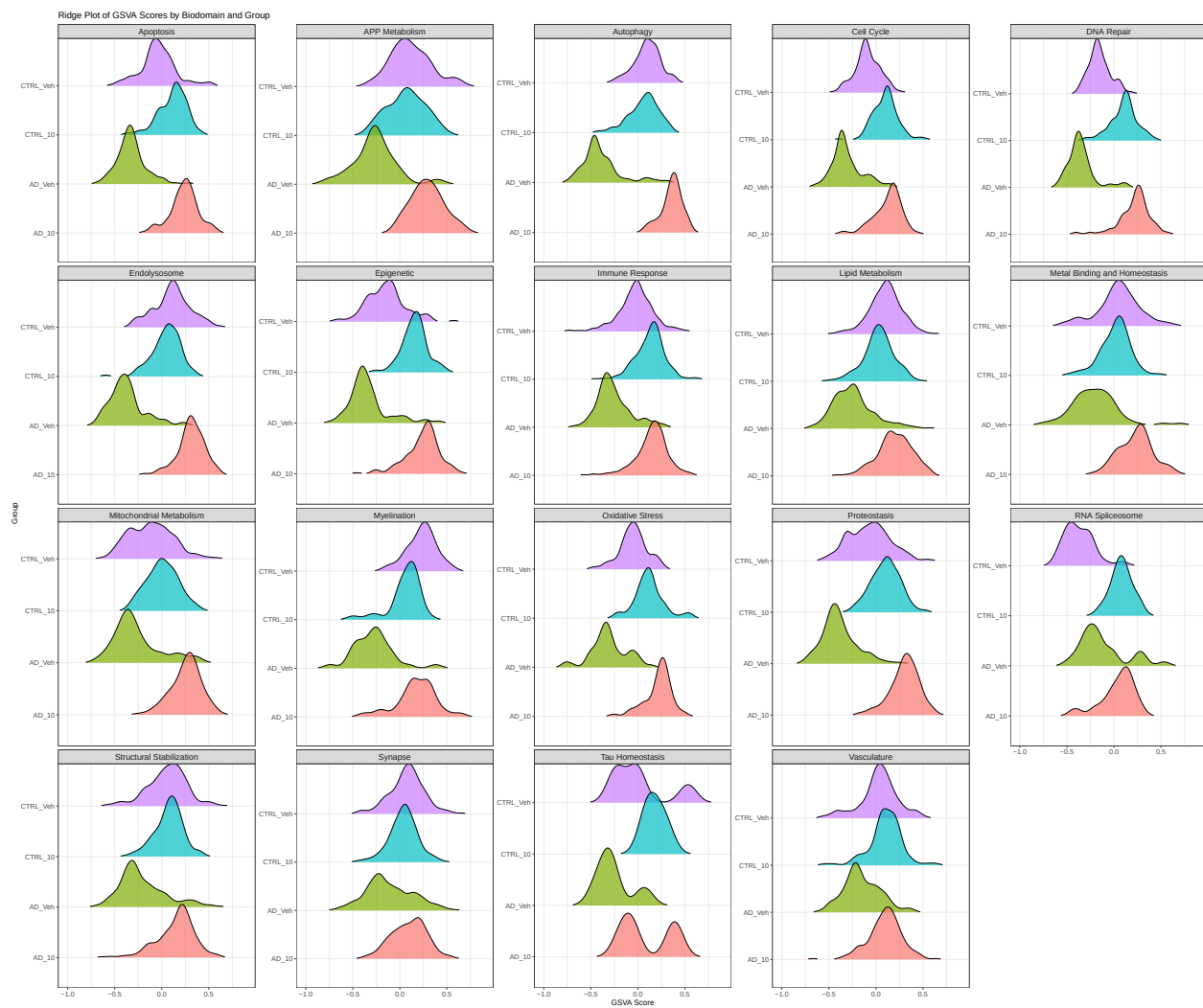
4. BioDomain analysis

5. Load the GSVA analysis

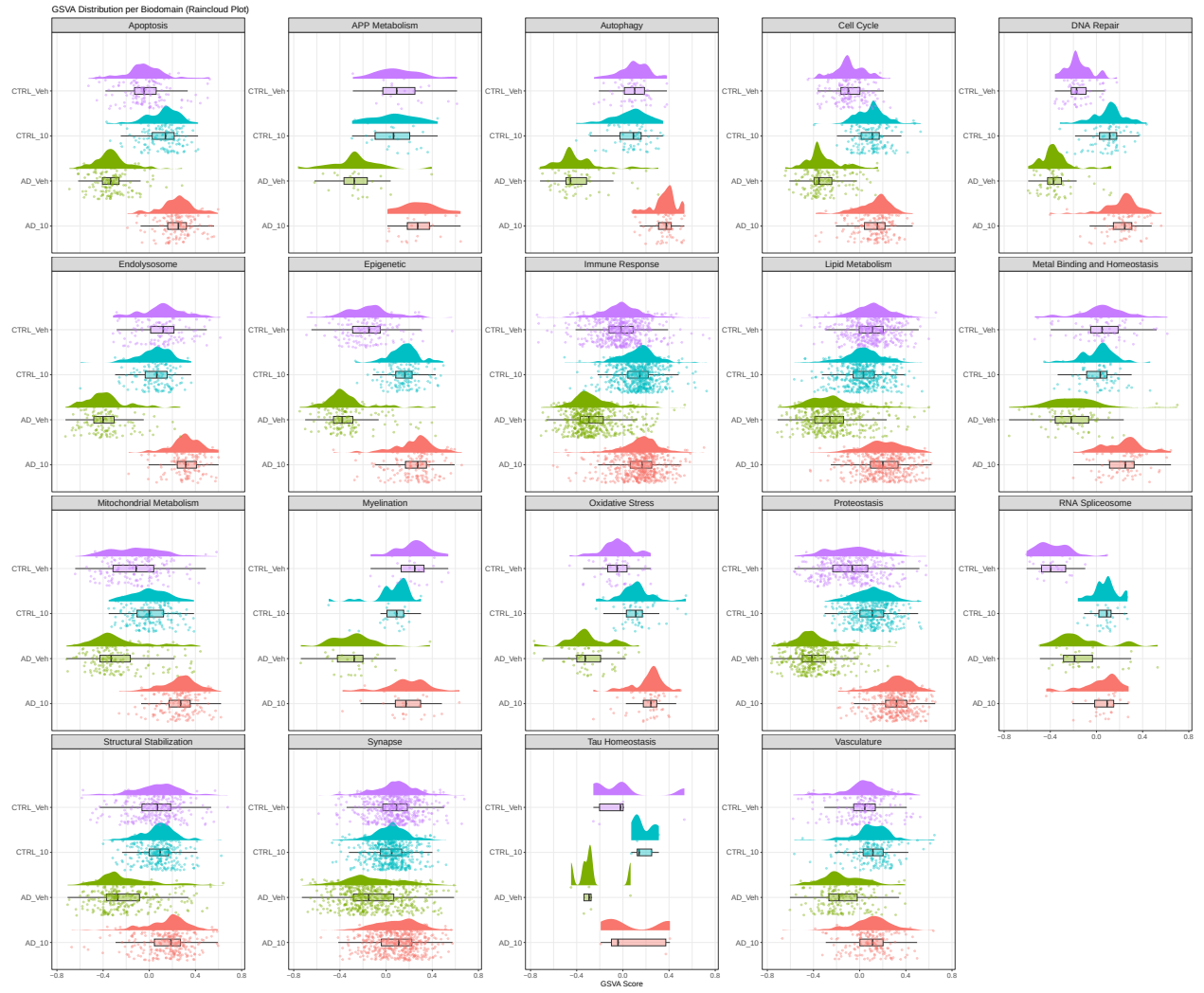
```
## [1] FALSE
## [1] 15598      9

##   gs_exact_source gs_cat gs_subcat
## 1      GO:0000002    C5      GO:BP
## 2      GO:0000003    C5      GO:BP
## 3      GO:0000012    C5      GO:BP
## 4      GO:0000014    C5      GO:MF
## 5      GO:0000018    C5      GO:BP
## 6      GO:0000019    C5      GO:BP
##
##                                     gs_name
## 1                      GOBP_MITOCHONDRIAL_GENOME_MAINTENANCE
## 2                      GOBP_REPRODUCTION
## 3                      GOBP_SINGLE_STRAND_BREAK_REPAIR
## 4 GOMF_SINGLE_STRANDED_DNA_ENDODEOXYRIBONUCLEASE_ACTIVITY
## 5                      GOBP_REGULATION_OF_DNA_RECOMBINATION
## 6                      GOBP_REGULATION_OF_MITOTIC_RECOMBINATION
##
##      Biodomain                                     G0term_Name
## 1 Mitochondrial Metabolism                      mitochondrial genome maintenance
## 2                      <NA>                                     <NA>
## 3                      DNA Repair                      single strand break repair
## 4                      DNA Repair single-stranded DNA endodeoxyribonuclease activity
## 5                      Cell Cycle                      regulation of DNA recombination
## 6                      <NA>                                     <NA>
##   n_ensGene n_entrezGene n_hgncSymbol
## 1         12          11          11
## 2         NA           NA           NA
## 3         11          11          11
## 4          9           9           9
## 5          5           5           5
## 6         NA           NA           NA
## [1] 11979
```

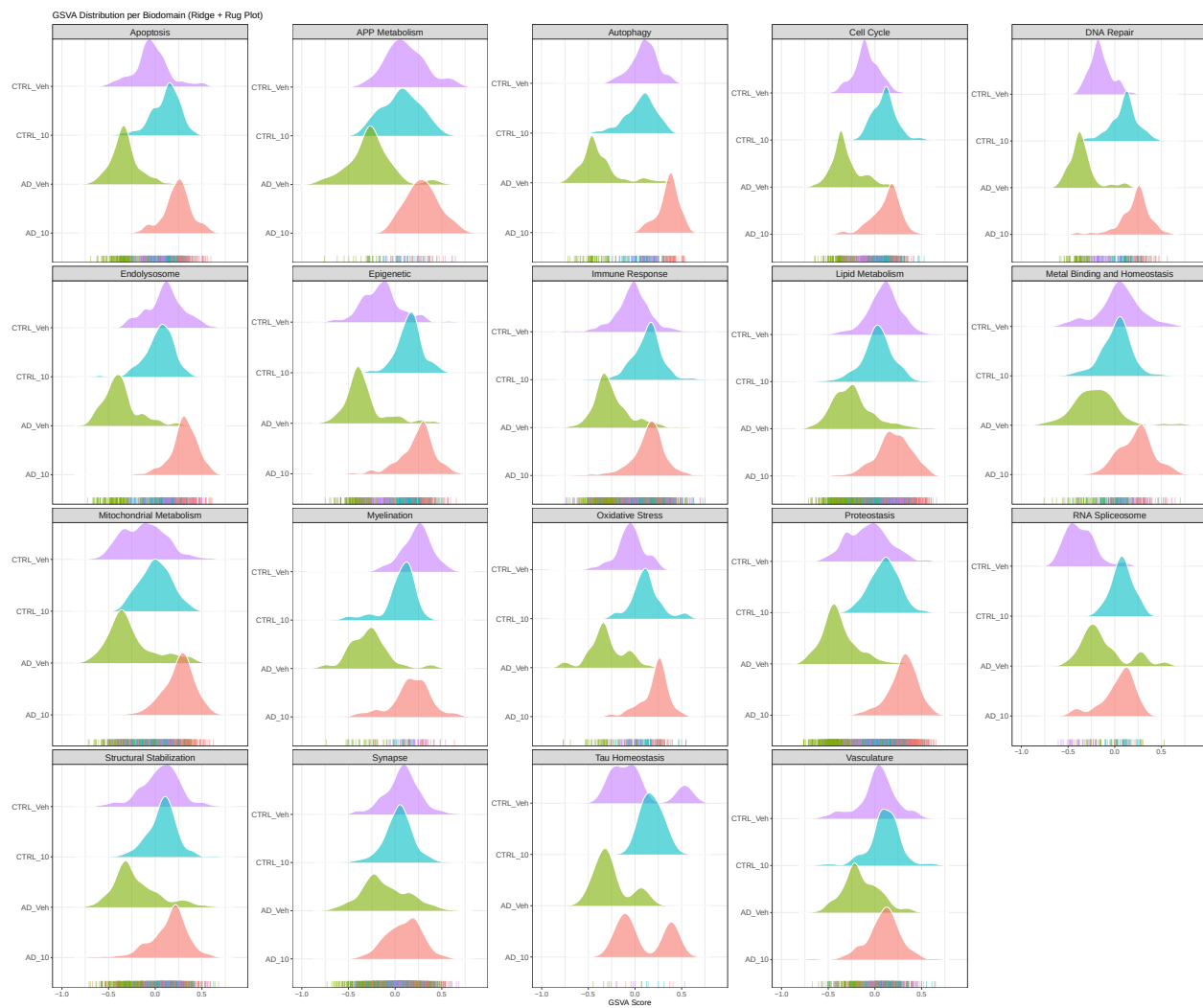
(1) Ridge plot for GSVA score



(2) Raincloud plot for GSVA score



(3) Rug and density plot for GSVA score



```

# Session information

## R version 4.4.0 (2024-04-24)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sonoma 14.3.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] parallel stats4 stats graphics grDevices utils datasets
## [8] methods base
##
## other attached packages:
## [1] ggdist_3.3.2 ggribes_0.5.6
## [3] GSEABase_1.66.0 graph_1.82.0
## [5] annotate_1.82.0 XML_3.99-0.18
## [7] extrafont_0.19 ggsignif_0.6.4
## [9] patchwork_1.3.0 decoupleR_2.10.0
## [11] GSVA_1.52.3 BiocParallel_1.38.0
## [13] edgeR_4.2.2 limma_3.60.6
## [15] GenomicFeatures_1.56.0 biomaRt_2.60.1
## [17] gprofiler2_0.2.3 RColorBrewer_1.1-3
## [19] data.table_1.16.4 org.Hs.eg.db_3.19.1
## [21] AnnotationDbi_1.66.0 clusterProfiler_4.12.6
## [23] ggfortify_0.4.17 pheatmap_1.0.12
## [25] EnhancedVolcano_1.22.0 ggrepel_0.9.6
## [27] apeglm_1.26.1 DESeq2_1.44.0
## [29] SummarizedExperiment_1.34.0 Biobase_2.64.0
## [31] MatrixGenerics_1.16.0 matrixStats_1.5.0
## [33] reshape2_1.4.4 Matrix_1.7-2
## [35] Signac_1.14.0 Seurat_5.2.1
## [37] SeuratObject_5.0.2 sp_2.2-0
## [39] rtracklayer_1.64.0 GenomicRanges_1.56.2
## [41] GenomeInfoDb_1.40.1 IRanges_2.38.1
## [43] S4Vectors_0.42.1 BiocGenerics_0.50.0
## [45] knitr_1.49 lubridate_1.9.4
## [47] forcats_1.0.0 stringr_1.5.1
## [49] dplyr_1.1.4 purrr_1.0.4
## [51] readr_2.1.5 tidyr_1.3.1
## [53] tibble_3.2.1 ggplot2_3.5.1
## [55] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] SpatialExperiment_1.14.0 R.methodsS3_1.8.2
## [3] progress_1.2.3 goftest_1.2-3
## [5] HDF5Array_1.32.1 Biostrings_2.72.1

```

##	[7]	vctr_0.6.5	spatstat.random_3.3-2
##	[9]	digest_0.6.37	png_0.1-8
##	[11]	deldir_2.0-4	parallelly_1.42.0
##	[13]	magick_2.8.5	MASS_7.3-64
##	[15]	httpuv_1.6.15	qvalue_2.36.0
##	[17]	withr_3.0.2	xfun_0.51
##	[19]	ggfun_0.1.8	survival_3.8-3
##	[21]	memoise_2.0.1	gson_0.1.0
##	[23]	systemfonts_1.2.1	ragg_1.3.3
##	[25]	tidytree_0.4.6	zoo_1.8-12
##	[27]	pbapply_1.7-2	R.oo_1.27.0
##	[29]	prettyunits_1.2.0	KEGGREST_1.44.1
##	[31]	promises_1.3.2	httr_1.4.7
##	[33]	restfulr_0.0.15	rhdf5filters_1.16.0
##	[35]	globals_0.16.3	fitdistrplus_1.2-2
##	[37]	rhdf5_2.48.0	rstudioapi_0.17.1
##	[39]	UCSC.utils_1.0.0	miniUI_0.1.1.1
##	[41]	generics_0.1.3	DOSE_3.30.5
##	[43]	curl_6.2.1	zlibbioc_1.50.0
##	[45]	ScaledMatrix_1.12.0	ggraph_2.2.1
##	[47]	polycip_1.10-7	GenomeInfoDbData_1.2.12
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##	[51]	evaluate_1.0.3	S4Arrays_1.4.1
##	[53]	BiocFileCache_2.12.0	hms_1.1.3
##	[55]	irlba_2.3.5.1	colorspace_2.1-1
##	[57]	filelock_1.0.3	ROCR_1.0-11
##	[59]	reticulate_1.40.0	spatstat.data_3.1-4
##	[61]	magrittr_2.0.3	lmtest_0.9-40
##	[63]	later_1.4.1	viridis_0.6.5
##	[65]	ggtree_3.12.0	lattice_0.22-6
##	[67]	spatstat.geom_3.3-5	future.apply_1.11.3
##	[69]	scattermore_1.2	shadowtext_0.1.4
##	[71]	cowplot_1.1.3	RcppAnnoy_0.0.22
##	[73]	pillar_1.10.1	nlme_3.1-167
##	[75]	compiler_4.4.0	beachmat_2.20.0
##	[77]	RSpectra_0.16-2	stringi_1.8.4
##	[79]	tensor_1.5	GenomicAlignments_1.40.0
##	[81]	plyr_1.8.9	crayon_1.5.3
##	[83]	abind_1.4-8	BiocIO_1.14.0
##	[85]	gridGraphics_0.5-1	emdbook_1.3.13
##	[87]	locfit_1.5-9.11	graphlayouts_1.2.2
##	[89]	bit_4.5.0.1	fastmatch_1.1-6
##	[91]	textshaping_1.0.0	codetools_0.2-20
##	[93]	BiocSingular_1.20.0	plotly_4.10.4
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##	[113]	RSQLite_2.3.9	viridisLite_0.4.2

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## [155] GO.db_3.19.1	sctransform_0.4.1
## [157] rappdirs_0.3.3	tinytex_0.55
## [159] glue_1.8.0	spam_2.11-1
## [161] httr2_1.1.0	XVector_0.44.0
## [163] RCurl_1.98-1.16	rprojroot_2.0.4
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## [167] extrafontdb_1.0	igraph_2.1.4
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